

Getting Started: Derivations

The HELP facility is *not* intended to provide help with the substance of logic. You can search this document to FIND a particular word or phrase by pressing Ctrl+F.

You may wish to print this document and have it before you as you begin doing derivations.

LOGIC NOTES:

- 1) In addition to the many derivations illustrated in the main part of the TerryText, there are many examples of solutions to exercises in the appendices to each chapter. LOGIC 2010 also includes examples of solutions to certain exercises. Such problems are marked “EG” (as in e.g. *for example*).
- 2) Advice on strategies for constructing derivations can be accessed from the Advice button. The TerryText also contain sections in Chapters 1, 2, and 3 on strategies. Note that it uses “available line” where we use “antecedent line” (or sometimes “available antecedent line”).
- 3) The official conditions that determine what constitutes a correct (unabbreviated) derivation for Chapters 1 and 2 can be found in the TerryText Chapter 2 Section 9.

KEYBOARD SHORTCUTS: Many buttons, including all those for logic symbols in the Toolboxes, have keyboard shortcuts. If you hold the mouse above such a button, a message containing the keyboard shortcut will appear. You should learn the keyboard shortcuts for the symbols you use frequently. A complete list is given in the Help document.

SELECTING FROM A LIST: To select more than one item from the Print or Submit list, use Shift+Click to select a string of adjacent problems, Ctrl+Click to select (or deselect) non-adjacent problems.

Within the Derivations module

When you select the Derivations Module, a blue screen should appear with the word “Problem:” written at the top in yellow and a double row of buttons at the bottom.

Selecting a problem to work on

Problems are stored in your *work* folder (L2KX\work) on your computer. Click the “Select” button at the bottom of the program window. This will bring up the list of problems to choose from with the preceding choice highlighted. Problems not yet attempted appear in BLACK. Problems already completed correctly appear in GREEN. Problems containing incorrect work appear in RED. To select a problem to work on, double-click it.

You can jump to a specific problem by typing in its number.

If you wish to print a list of problems for review, click Print (in the button bar at the bottom of the screen), select the problems you wish to print (see [Selecting From a List](#) above) and then click Print List.

Examples

Examples of completed derivations are provided to illustrate derivational techniques. All Examples have names ending in “EG”.

Creating your own ‘User’ problem

Instead of selecting one of the preset problems, you may wish to work on a problem of your own creation or a problem that your instructor has proposed but that has not been programmed into the Select list. Such problems are called “User” problems. To create a new User problem, click the User button on the Derivation screen, and enter the problem. To enter the logic symbols, either right-click in the Problem field to invoke the Toolbox and use the buttons containing the symbols or use the [Keyboard Shortcuts](#).

In creating a new problem, remember that each premise, except the last, must be followed by a period, and that the “therefore” symbol, **Ctrl+Shift+T** (or use the Toolbox button), must appear between the last premise and the conclusion.

User problems are saved to the bottom of the list of exercises. When you save a User problem, you will be prompted to enter a name. The prefix “User” will be suggested, but almost any name will work.

Starting a Derivation

After selecting an exercise or creating a User problem, press Enter to produce the first line of your derivation. Lines are numbered automatically. If you delete or insert a line, the lines are renumbered automatically (as are the line references in the justifications).

Show lines

To enter a Show line simply type “Show”, followed by a space, at the beginning of the line. Show lines do not require a justification.

Non-Show lines

For each non-Show line, you provide a formula and, to the right, a justification. Use the Tab key to move back and forth between the WFF field and the Justification field. The program includes a set of editing tools, including Toolboxes appropriate to each field. A Toolbox is always invoked by a right-click of the mouse in the given field. Anything you type while a Toolbox is open will be passed through to the line. So you can use a combination of typing and clicking on Toolbox buttons.

In a typical justification, which invokes a rule and refers to available antecedent lines, e.g. "3 7 MP", the line references must precede the abbreviated rule name, and each of the line references and the rule name must be separated from one another by a space or other separator.

Command Mode

LOGIC 2010 allows you to enter lines using Command Mode. In Command Mode, you enter the justification, without entering the line, press **Enter**, and the program will enter the line for you (or explain in an error message why the command could not be executed). Command Mode ordinarily creates a new line awaiting a new command. To avoid creating a new line (for example, at the last line of a derivation) use **Shift+Enter** to execute the command.

Show Commands

The Show Commands automate the entry of Show lines. They are keyed to the "Brief Strategic Advice" document accessible from the Advice button.

For more on the tools and techniques available for constructing derivations see the main Help document for this module.

Checking your work

You can check your work at any time by clicking the Check button at the bottom of the Derivation screen (or use **Ctrl+k**). If your derivation is correct, the message "Correct Derivation" will appear in the upper right hand corner as part of the Problem Line.

If your derivation contains errors, a message containing a brief description of the error will appear on each erroneous line. Further explanations can be invoked by clicking the  button next to the message. Not all messages indicate errors, only those with a red background; those with a blue background simply provide information.

Even if you correct the errors, the messages will remain until you again check the work. Error messages are not saved with your work, so you will need to recheck your work when a problem that has been saved is recalled.

Saving your work

Save your finished or unfinished work on a problem by clicking the "Save" button at the bottom of the Derivation screen (or use **Ctrl+S**). This will save the current version of your work to the *work* directory on your computer.

Printing your work

You can print your work at any time by clicking the Print button at the bottom of the screen. (Note: Some computer labs may require you to open a "print account" at the lab.) You may select more than one exercise to print. Use

Ctrl+Click to select additional problems. Choose Print to print the full derivations, one to a page, or choose Print Results to print only results. If you choose to Print Results, each of the exercises you have selected will be represented by a single line, which will contain the problem and an indication of whether the exercise was completed correctly or not. Usually only a single page is used for all results.

Submitting your work

You can also submit your work over the Internet to the database of homework submissions established for your course. Click the Submit button at the bottom of the screen. You will be asked to verify course information, and to enter your logic password, which you will have created when you Registered with LOGIC 2010. Just as with printing, you may select more than one exercise to submit. Wait for the program to verify that the submitted work has been received.

You should check your work before submitting it. In general, it is better not to submit incorrect work. You can submit more than one version of a given problem. *The last version submitted before the due date of a homework assignment is the one that counts.*

Justifications for boxing and canceling

Form of B&C	Justification
	Note that line numbers must precede the rule name.
Direct derivation	line that matches the show line + DD
Conditional derivation	line containing the consequent of the show line + CD
Indirect derivation	two lines that contradict each other + ID
Universal derivation	line that matches the show line, but with the initial universal quantifier phrase removed + UD

The lines you cite for boxing and canceling must appear on available antecedent lines below the show line you are canceling.

Button Bar buttons:

Button	Function
Select (Select Problem) Ctrl+o	Used to select a problem to work on. You may select one of the preset exercises or a User problem that you have already created and saved.
User (User Problem)	Used to create a "User Problem", a new problem of your own that is not in the list of exercises.
Rules (Display Available Rules)	Pops up a window showing all the inference rules. For derived rules, the theorems necessary to derive the rule are indicated.
Check Ctrl+k	Initiates an automated check of your work on the current problem, and displays messages describing any errors.
Save Ctrl+s	Saves your work on the current problem to your work folder, where it will be available for you to recall it for editing. When

Button	Function
	you leave the Workspace you will be prompted to save.
Delete	Pops up a window asking if you wish to delete the work on the current problem (and begin anew). If the problem is a User problem, you also have the option of deleting the problem itself.
Submit	Submit one or several problems over the Internet to the course database. After clicking the Submit button you will verify your course, instructor, and TA, enter your logic password, and select the exercises you wish to submit using Ctrl+Click . Then click Submit at the bottom of the selection window to submit all of the selected problems. After a brief delay, you will receive an acknowledgment that the work has been successfully received.
Print Ctrl+p	Pops up the Print window, which lists all of the preset exercises along with all of the User problems that you have created and saved. The current problem will be highlighted. Use Ctrl+Click to select (or deselect) more than one exercise to print. The Print button at the bottom will then initiate printing of your work, one problem per page. You may also choose the Print Results button, which causes a single page to be printed showing just a list of the selected problems with the results (Correct or Incorrect) of your work, but without showing the work itself. You can also print a list of exercises (without work) by using the Print List button.
Menu	Summon the Menu window.
Close	Close the module window. You will be asked if you wish to save your current work before the window is closed.
Feedback	Send anonymous suggestions about the program or any aspect of the course to the instructor, TAs, and the program developers. Questions and comments requiring an answer should include you name and email address.
Starting	Summon this document.
Help	Read a full explanation of how the module works, including shortcut derivation tools.
FAQ	Read the answers to Frequently Asked Questions.
Advice	Summon the "Brief Strategic Advice" document. The Advice for Chapter II is different from the Advice for Chapter I.

Keystrokes

Use...	To...
Enter	Produces a new line (and, if in Command Mode, executes the command).
Shift+Enter	Executes a command in Command Mode without producing a new line.
Ctrl+Shift+T	Produces a 'Therefore' sign when manually entering a User problem.
Ctrl+Shift+S	Enters a <u>S</u> how field (or removes one that is present)

Ctrl+Shift+C	Produces a <u>C</u> onditional sign (an arrow)
Ctrl+Shift+N	Produces a <u>N</u> egation sign (a tilde)
Alt+Delete	<u>D</u> eletes the current line
Ctrl+K	Check <u>K</u> s your work
Ctrl+Shift+X	Bo <u>X</u> es and Cancels (or removes an existing box)

Show Commands

Show Commands allow one to enter the most commonly used Show lines while in Command Mode. The Show Commands are keyed to the strategies suggested in the “Strategic Advice” document.

For each Show Command, it suffices to type a sufficient initial portion of the command to identify it. Thus, for the command “Show Conclusion”, any of the following will suffice: “Show Conc”, “Show Concl”, “Show Conclu”, “Show Conclus”, “Show Conclusi”, and so on. In the following, brackets [] are used to show the optional part of the command.

Show Commands are entered into the Justification field (as are all Command Mode commands) and cannot be combined with (i.e. queued with) any other command.

To undo a Show Command, put the cursor in the Show line (not the Justification field), and press **Ctrl+Shift+S**, or right-click to bring up the Toolbox, and then click Show/Unshow.

For Chapter I, the relevant commands are:

Show Command	Result
Show Conc[clusion]	Used to enter the initial line of a derivation. (Strategy 1.)
Show Cons[sequent]	When the immediately preceding uncanceled Show line is a conditional, and you have already used ASS CD to assume the antecedent, use this command to enter a Show line for the Consequent. (Strategy 4B.)
Show Unneg[ation] n	When available antecedent line n is the negation of a conditional, and you have already begun an indirect derivation (by using ASS ID), use this command to enter a Show line for the Conditional. (Strategy 8.)
Show Ant[ecedent] n	When available antecedent line n is a conditional, and neither MP nor MT can be applied directly, use this command to enter a Show line for the antecedent of the conditional. (Strategy 9.)